



ADAPTIVE SMART LIGHTS: REDEFINING ENERGY CONSUMPTION

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ABSTRACT

With rapid urbanization and the increasing demand for sustainable energy solutions, traditional street lighting systems pose significant challenges in terms of energy consumption and operational costs. Conventional streetlights remain operational at full intensity throughout the night, leading to unnecessary energy wastage and increased carbon emissions. The advent of smart cities and the integration of Internet of Things (IoT) technologies provide an opportunity to develop intelligent, adaptive lighting solutions that optimize energy usage while maintaining road safety and visibility.

This study aims to design and implement a Smart Adaptive Street Lighting System that dynamically adjusts brightness levels based on real-time vehicular and pedestrian movement. The goal is to enhance energy efficiency, reduce operational costs, and promote sustainable urban development without compromising public safety.

A Python-based simulation was developed using Pygame for real-time traffic movement and Streamlit for a dashboard interface displaying streetlight status and energy consumption. The adaptive lighting mechanism was programmed to adjust illumination levels based on the proximity of moving vehicles. The system was evaluated by measuring the reduction in energy consumption and responsiveness to varying traffic conditions.

The proposed smart lighting system successfully demonstrated a reduction in energy consumption by dynamically adjusting brightness levels. Streetlights near active traffic remained illuminated at optimal intensity, while those in low-traffic zones dimmed significantly, reducing unnecessary power usage. The simulation results indicate that the adaptive system can lower energy consumption by up to 40%, offering a viable solution for municipalities aiming for sustainable infrastructure.

The study highlights the potential of adaptive street lighting in revolutionizing urban energy management. By integrating real-time traffic sensing and adaptive control, cities can significantly cut energy costs, reduce carbon footprints, and extend streetlight lifespan. Future research can explore IoT-based implementations and real-world deployment to further refine and enhance system efficiency.

INTRODUCTION

Context and Background

As urbanization accelerates, cities worldwide are seeking innovative solutions to optimize resource consumption while maintaining infrastructure efficiency. Among the most energy-intensive urban facilities is street lighting, which accounts for a significant portion of municipal electricity usage, often exceeding 30-40% of a city's total energy budget. Traditional street lighting systems operate on fixed schedules, remaining at full brightness throughout the night, regardless of pedestrian or vehicular activity. This results in unnecessary energy consumption, higher maintenance costs, and increased carbon emissions.

With the emergence of smart cities and Internet of Things (IoT) technologies, adaptive street lighting presents an opportunity to enhance energy efficiency through real-time sensing and automation. Smart streetlights can intelligently adjust brightness levels based on surrounding conditions, ensuring that energy is used only when necessary while maintaining public safety.

Problem Statement

Despite technological advancements, most urban street lighting systems continue to follow static, pre-programmed schedules, leading to significant energy wastage during low-traffic hours. Existing solutions often rely on manual control or outdated sensors with limited adaptability to real-time traffic conditions. This inefficiency translates into:

- Unnecessary energy consumption during non-peak hours.
- Higher maintenance costs due to prolonged full-intensity operation.
- Increased carbon footprint, contradicting global sustainability goals.

Importance of the Study

The transition to smart, adaptive lighting is crucial in reducing energy consumption and promoting environmental sustainability. Cities can cut costs, extend infrastructure lifespan, and align with energy conservation policies by implementing automated, responsive streetlight systems. Given the increasing push toward carbon-neutral urban planning, the need for such intelligent lighting solutions has never been more urgent.

Several studies have explored energy-efficient street lighting solutions:

- IoT-based Smart Streetlights: Research has demonstrated the viability of integrating IoT-based adaptive lighting systems using sensors and cloud networks to regulate brightness dynamically.
- Motion-Sensing Streetlights: Prior studies have incorporated motion sensors to adjust brightness levels based on pedestrian and vehicle movement, achieving partial success in energy conservation.
- AI & Predictive Algorithms: Machine learning-based predictive lighting models have been proposed to optimize lighting based on historical traffic patterns.

While these studies highlight promising developments, most existing implementations require complex hardware integration or rely on expensive infrastructure upgrades, limiting their widespread adoption.

Research Gap

While adaptive lighting solutions exist, affordable and easily deployable models remain underdeveloped. Current solutions often:

- Require expensive sensor installations and high infrastructural modifications.
- Lack efficient, real-time control mechanisms that dynamically adjust brightness based on live traffic flow.
- Are not well-integrated with digital monitoring systems for visualization and management.

Gaps Addressed

This research proposes a software-driven, simulation-based approach to adaptive street lighting, demonstrating a cost-effective, scalable solution that can be implemented with minimal hardware dependencies. Key contributions of this study include:

- A Python-based smart lighting simulation, integrating Pygame for real-time traffic visualization and Streamlit for a monitoring dashboard.
- An adaptive control mechanism that dynamically adjusts brightness based on vehicle movement, reducing energy wastage.
- A real-time energy consumption tracking system, offering a practical framework for evaluating efficiency improvements.

By addressing the challenges of cost and real-time control, this study bridges the gap between theoretical models and practical, deployable solutions for smart city infrastructure. The findings from this research could serve as a foundation for future real-world implementations, supporting sustainable urban development.

LITERATURE SURVEY

An Energy-efficient Smart Street Lighting System with Adaptive Control based on Environment: Nowadays, most street lighting consumes high electrical power due to inefficient design. Energy efficiency and ease of maintenance with minimal involvement of manual work in street lights management are two top priorities of a smart city. To fulfil these purposes, the project aims to design and develop a smart street lighting system based on wireless sensing network technology and a novel adaptive control mechanism for the benefits of energy-saving and ease of maintenance. Experimental results show up to 48% potential energy saving compared to a conventional street lighting system.

Reviewing the parameters that affect sustainability and energy efficient concepts in road and urban lighting design: This study systematically reviews energy management in street lighting, examining key factors such as design parameters, adaptive lighting, and technological advancements. By identifying research gaps, it aims to guide future studies toward more efficient and sustainable urban lighting solutions. This study aims to analyze critical design factors in road lighting, bridging theoretical principles with real-world applications to maximize energy savings and sustainability.

Advanced Adaptive Street Lighting Systems for Smart Cities: This paper presents the findings of the SCALS (Smart Cities Adaptive Lighting System) project, which developed a cost-effective smart street lighting system. The system enables municipalities to manage and control public street lamps, autonomously adjusting brightness based on the presence of vehicles or pedestrians. This adaptive approach significantly reduces energy consumption while integrating into a broader IoT infrastructure, where each lighting pole functions as a network

element. The smart system incorporates various technologies, including motion sensors, video cameras, weather sensors, and remote management tools, to optimize lighting efficiency.

Adaptive Street Lighting Predictive Control: In this paper an implementation of a smart predictive monitoring and adaptive control system for the public lighting have been carried out. The vehicular traffic flow acquired using a smart camera has been analyzed and several predictive methods have been studied. Then, a control strategy based on the given traffic forecasts and on the dynamical street class downgrade allowed by the law, has been implemented. Experimental results provided by a real life testbed showed that the proposed strategy has high potential energy savings without affecting safety.

Energy Saving Potential of Traffic-Regulated Street Lighting: To combat rising electricity costs, some municipalities in Europe turn off streetlights for several hours at night, raising safety concerns. The EN 13201 standard allows for reducing road luminance based on traffic volume, balancing energy savings with safety. This study proposes a method to estimate energy savings from traffic-regulated street lighting using one year of traffic data from an urban street. Findings show that sensor-based lighting adjustments can reduce electricity costs by up to 55% while maintaining compliance with EN 13201-2. Additionally, a pre-programmed dimming schedule could achieve 45% energy savings. The study also examines how daylight-saving time impacts adaptive street lighting efficiency.

PROPOSED METHODOLOGY

This project focuses on the simulation and visualization of a smart adaptive lighting system using Python-based simulation (Pygame) and a real-time monitoring dashboard (Streamlit). The approach models streetlights that dynamically adjust brightness based on the movement of vehicles, aiming to optimize energy efficiency while maintaining adequate illumination.

1. Framework and Techniques Used

1.1 Frameworks and Libraries

- Pygame: Used for simulating the street environment, including vehicles and dynamic streetlights.
- Streamlit: Provides an interactive dashboard to display real-time streetlight status and energy consumption.
- Threading: Ensures that the simulation and dashboard run simultaneously without blocking each other.
- Time-based Updating: The brightness of streetlights adjusts dynamically based on vehicle proximity using a smooth transition model.

1.2 Techniques Implemented

- Object-oriented modeling for vehicles and streetlights.
- Distance-based brightness control: Light intensity is adjusted based on vehicle distance from each streetlight.
- Energy Consumption Estimation: The dashboard continuously updates the estimated energy usage based on light intensity.

2. Step-by-Step Approach

Step 1: Simulation of Smart Streetlights and Vehicles

- The Pygame module creates a virtual street with multiple lanes.
- Vehicles are randomly spawned and move across the screen at varying speeds.
- Streetlights are positioned at regular intervals along the road.

Step 2: Adaptive Brightness Adjustment Algorithm

- Each streetlight checks the distance to the nearest vehicle.
- If a vehicle is within a threshold (e.g., 100 pixels), the brightness increases to full intensity.
- If no vehicle is nearby, the brightness gradually dims to a lower level.

- A smooth transition function ensures that changes in brightness do not appear abrupt.

Step 3: Energy Consumption Estimation

- The energy consumption of each streetlight is estimated using the formula:

$$E = \frac{I}{I_{max}} \times P$$

where:

- E = Energy consumption per streetlight (arbitrary unit)
- I = Current brightness level of the streetlight
- I_{max} = Maximum brightness level
- P = Maximum power usage per streetlight

Step 4: Real-Time Dashboard Updates

- The Streamlit dashboard continuously retrieves and displays:
 - The current status of each streetlight (ON/OFF).
 - The corresponding energy consumption for each light.
- A multi-threading approach ensures that the simulation and dashboard run in parallel without lag.

3. System Architecture and Process Flow

3.1 System Architecture

The system consists of two main components:

1. Simulation Module (Pygame): Handles vehicle movement, streetlight brightness control, and energy calculation.
2. Dashboard Module (Streamlit): Displays real-time data on streetlight status and energy consumption.

3.2 Process Flow

1. Simulation Starts: Vehicles move along a predefined road.

2. Streetlight Activation: Lights adjust brightness based on the vehicle's position.
3. Energy Computation: The system calculates the total energy consumption dynamically.
4. Data Transfer to Dashboard: The energy values and light status are updated in real-time on the Streamlit dashboard.

4. Model Parameters and Assumptions

Parameter	Description	Value (Example)	
Road Width	Number of lanes in the simulated road	2 Lanes	
Streetlight Spacing	Distance between two streetlights	200 pixels	
Vehicle Speed	Randomized speed for each vehicle	2-4 pixels/frame	
Brightness Levels	Adaptive brightness levels	100 (dim) to 255 (bright)	
Detection Range	Distance within which a streetlight reacts to a vehicle	100 pixels	
Energy Calculation	Assumed power unit for simulation purposes	10 units per full brightness	

Assumptions:

- Vehicles move in a straight path without sudden stops.
- The brightness change is gradual to prevent flickering.
- The system assumes equal energy distribution among all streetlights.
- The simulation does not consider weather conditions or external lighting.

EXPERIMENTATION

Simulation Environment

The project employs a Python-based simulation using Pygame to model the adaptive lighting system in a controlled virtual environment. The simulation replicates an urban street scenario with moving vehicles and dynamically adjusting streetlights based on vehicle proximity.

Dashboard Implementation

A Streamlit-based dashboard is integrated with the simulation to display real-time streetlight status and energy consumption. The dashboard provides an intuitive interface for monitoring lighting conditions and energy efficiency.

Data Collection and Processing

- **Vehicle Movement Data:** The simulation generates vehicle movements with random speeds and positions.
- **Streetlight Adaptation:** The lighting intensity is adjusted based on the proximity of vehicles to streetlight positions.
- **Energy Consumption Calculation:** The system estimates energy usage dynamically, considering the brightness level of each streetlight.

Evaluation Metrics

- **Streetlight Response Time:** Measures the latency between vehicle detection and streetlight brightness adjustment.
- **Energy Savings Estimation:** Compares the adaptive lighting system's energy usage with a constant-intensity traditional lighting approach.
- **Simulation Accuracy:** Ensures smooth transitions and realistic streetlight behavior through fine-tuned thresholds.

Software Tools Used

- **Python Libraries:**
 - Pygame (for simulation and vehicle movements)
 - Streamlit (for dashboard visualization)
 - Threading and Time (for real-time updates)
- **Development Environment:** Visual Studio Code (VS Code)

The experimentation phase successfully demonstrates the impact of an adaptive lighting system without requiring real-world hardware deployment.

RESULTS AND DISCUSSION

DASHBOARD CODE:

```
Dashboard.py > ...
1  import pygame
2  import streamlit as st
3  import random
4  import time
5  import threading
6
7  # Initialize Pygame
8  def run_simulation(shared_data):
9      pygame.init()
10     WIDTH, HEIGHT = 800, 400
11     screen = pygame.display.set_mode((WIDTH, HEIGHT))
12     pygame.display.set_caption("Smart Adapter Lights Simulation")
13
14     BLACK = (0, 0, 0)
15     GRAY = (100, 100, 100)
16     WHITE = (255, 255, 255)
17
18     road_y = HEIGHT // 2
19     lanes = [road_y - 40, road_y + 20]
20     streetlights = [(150, road_y - 70), (350, road_y - 70), (550, road_y - 70), (750, road_y - 70)]
21
22
23     class Vehicle:
24         def __init__(self, x, y, speed):
25             self.rect = pygame.Rect(x, y, 50, 30)
26             self.speed = speed
27
28         def move(self):
29             self.rect.x += self.speed
30             if self.rect.x > WIDTH:
31                 self.rect.x = -50
32
33     vehicles = [Vehicle(random.randint(-200, WIDTH), random.choice(lanes), random.randint(2, 4)) for _ in range(3)]
34     light_intensity = {pos: 100 for pos in streetlights}
35     running = True
36     clock = pygame.time.Clock()
37
38     while running:
39         screen.fill(GRAY)
40         pygame.draw.rect(screen, BLACK, (0, road_y - 50, WIDTH, 100))
41
42         for vehicle in vehicles:
43             vehicle.move()
```

```

43     pygame.draw.rect(screen, WHITE, vehicle.rect)
44     total_energy = 0
45     for (x, y) in streetlights:
46         min_distance = min(abs(vehicle.rect.x - x) for vehicle in vehicles)
47         target_brightness = 255 if min_distance < 100 else 100
48         light_intensity[(x, y)] += (target_brightness - light_intensity[(x, y)]) * 0.1
49         light_color = (light_intensity[(x, y)], light_intensity[(x, y)], 0)
50         pygame.draw.circle(screen, light_color, (x, y), 20)
51
52         # Calculate energy consumption (arbitrary unit)
53         energy_usage = light_intensity[(x, y)] / 255 * 10
54         total_energy += energy_usage
55         shared_data[(x, y)] = ("ON" if light_intensity[(x, y)] > 150 else "OFF", round(energy_usage, 2))
56
57     for event in pygame.event.get():
58         if event.type == pygame.QUIT:
59             running = False
60
61     pygame.display.update()
62     clock.tick(30)
63
64     pygame.quit()
65
66 # Streamlit UI
67 def start_dashboard():
68     st.title("Smart Adapter Lights Dashboard")
69     st.write("This dashboard controls the smart lighting simulation.")
70     shared_data = {pos: ("OFF", 0) for pos in [(150, 330), (350, 330), (550, 330), (750, 330)]}
71
72     if st.button("Start Simulation"):
73         threading.Thread(target=run_simulation, args=(shared_data,), daemon=True).start()
74         time.sleep(1)
75         st.success("Simulation Running!")

```

```

79     while True:
80         status_text = "\n".join([f"Streetlight at {pos}: Status = {status}, Energy = {energy} units" for pos, (status, energy) in shared_data.items()])
81         status_placeholder.text(status_text)
82         time.sleep(1)
83
84     if __name__ == "__main__":
85         start_dashboard()
86

```

SIMULATION CODE:

```
Simulation.py > ...
1  import pygame
2  import random
3
4  # Initialize Pygame
5  pygame.init()
6
7  # Screen dimensions
8  WIDTH, HEIGHT = 800, 400
9  screen = pygame.display.set_mode((WIDTH, HEIGHT))
10 pygame.display.set_caption("Realistic Smart Adapter Lights Simulation")
11
12 # Colors
13 BLACK = (0, 0, 0)
14 GRAY = (100, 100, 100)
15 WHITE = (255, 255, 255)
16 YELLOW_DIM = (100, 100, 0)
17 YELLOW_BRIGHT = (255, 255, 0)
18
19 # Road settings
20 road_y = HEIGHT // 2
21 lanes = [road_y - 40, road_y + 20] # Two lanes
22 streetlights = [(150, road_y - 70), (350, road_y - 70), (550, road_y - 70), (750, road_y - 70)]
23
24 # Vehicle settings
25 class Vehicle:
26     def __init__(self, x, y, speed):
27         self.rect = pygame.Rect(x, y, 50, 30)
28         self.speed = speed
29
30     def move(self):
31         self.rect.x += self.speed
32         if self.rect.x > WIDTH:
33             self.rect.x = -50 # Reset position
34
35 # Generate multiple vehicles
36 vehicles = [Vehicle(random.randint(-200, WIDTH), random.choice(lanes), random.randint(2, 4)) for _ in range(3)]
37
38 # Light transition dictionary
39 light_intensity = {pos: 100 for pos in streetlights} # Default dim
40
41 # Main loop
42 running = True
43 clock = pygame.time.Clock()
44
45 while running:
46     screen.fill(GRAY)
47     pygame.draw.rect(screen, BLACK, (0, road_y - 50, WIDTH, 100)) # Road
48
49     # Move vehicles
50     for vehicle in vehicles:
51         vehicle.move()
52         pygame.draw.rect(screen, WHITE, vehicle.rect)
53
54     # Adaptive streetlight brightness
55     for (x, y) in streetlights:
56         min_distance = min(abs(vehicle.rect.x - x) for vehicle in vehicles) # Find closest vehicle
57         target_brightness = 255 if min_distance < 100 else 100 # Bright when close
58
59         # Smooth brightness transition
60         light_intensity[(x, y)] += (target_brightness - light_intensity[(x, y)]) * 0.1
61         light_color = (light_intensity[(x, y)], light_intensity[(x, y)], 0)
62
63         pygame.draw.circle(screen, light_color, (x, y), 20)
64
65     # Event handling
66     for event in pygame.event.get():
67         if event.type == pygame.QUIT:
68             running = False
69
70     pygame.display.update()
71     clock.tick(30) # 30 FPS for smooth movement
72
73 pygame.quit()
```

The project consists of two key components:

1. **Pygame Simulation** – Models the movement of vehicles and adjusts streetlight brightness dynamically.
2. **Streamlit Dashboard** – Provides a real-time visualization of the streetlight status and energy consumption.

1. Pygame Simulation (Smart Lighting System)

The Pygame simulation creates a virtual street environment where streetlights adjust their brightness based on vehicle movement.

Key Components:

- **Initialization:**
 - The simulation initializes a screen with a fixed width and height.
 - Colors are defined for road, vehicles, and streetlights.
 - Vehicles and streetlights are placed at predefined locations.
- **Vehicle Movement:**
 - Vehicles move along lanes at different speeds.
 - When a vehicle moves past a streetlight, the streetlight increases its brightness.
- **Adaptive Lighting Mechanism:**
 - The brightness of each streetlight is adjusted based on the nearest vehicle.
 - If a vehicle is close, the light intensity increases (bright yellow).
 - If no vehicles are nearby, the brightness decreases (dim yellow).
 - The transition in brightness is smooth using incremental updates.
- **Energy Consumption Calculation:**
 - Energy usage is computed based on brightness intensity.
 - A higher brightness level results in more energy consumption.

- **Real-Time Updates:**

- The Pygame loop continuously updates vehicle positions and light intensities.
- Events like quitting the simulation are handled.

2. Streamlit Dashboard (Real-Time Monitoring)

The Streamlit dashboard provides a visual representation of streetlight status and energy consumption.

Key Components:

- **Dashboard Initialization:**

- A Streamlit web app is created to display information.
- A shared data dictionary is used to store the state of each streetlight.

- **Simulation Control:**

- A "Start Simulation" button triggers the Pygame simulation in a separate thread.
- Once started, the simulation continuously updates streetlight status.

- **Real-Time Data Display:**

- The dashboard fetches the latest streetlight status from the shared dictionary.
- Each streetlight's "ON/OFF" status and energy consumption are displayed dynamically.
- The dashboard refreshes periodically to keep values updated.

How the Components Work Together

1. **User starts the simulation** from the Streamlit dashboard.
2. **Pygame runs the street simulation** with vehicles and adaptive streetlights.
3. **Streetlight brightness is updated** based on vehicle proximity.

4. **Energy consumption is calculated dynamically** in the Pygame loop.
5. **The updated streetlight status is shared** with the Streamlit dashboard.
6. **The dashboard refreshes** periodically to display the latest values.

CONCLUSIONS

This project successfully demonstrates an adaptive smart street lighting system through simulation and real-time monitoring using Pygame and Streamlit. The system dynamically adjusts streetlight brightness based on vehicle movement, optimizing energy consumption while ensuring adequate illumination. The dashboard provides an interactive and real-time visualization of streetlight status and energy usage, making it an effective tool for monitoring and analysis.

Key Takeaways:

1. **Energy Efficiency:** The adaptive lighting mechanism reduces unnecessary power consumption by dimming lights when no vehicles are detected.
2. **Real-Time Monitoring:** The Streamlit dashboard offers a user-friendly interface to track streetlight status and energy usage dynamically.
3. **Scalability & Future Applications:** The approach can be extended to real-world smart city infrastructures, integrating IoT-based sensors and AI-driven decision-making for improved efficiency.

This simulation-based study highlights the importance of smart lighting solutions in urban environments, contributing to sustainability by reducing energy waste while maintaining road safety. Future work can involve real-world implementation using IoT-enabled sensors, deep learning for traffic pattern analysis, and integration with renewable energy sources for enhanced sustainability.

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